BIWS – 4-Page Technical Cheat Sheet for Investment Banking Interviews

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Finance Concepts

Time Value of Money: Money today is worth **MORE** than money tomorrow because you could invest it today and earn more by tomorrow.

Present Value (PV) =

$$\frac{\text{Future Value}}{(1 + \text{Discount Rate})\ \hat{}\ \#\ \text{Years}}$$

Discount Rate: Measures both the risk and the potential returns of an investment; represents the annualized returns you're expecting or targeting on other, similar investments.

Net Present Value (NPV): Present Value – Upfront Cost or Asking Price.

IRR (Internal Rate of Return): The annualized return you expect to earn from *this specific investment*, regardless of what similar ones offer. Like the compound interest rate on a simple investment.

Invest When: IRR > Discount Rate or NPV > 0.
Do Not Invest When: IRR < Discount Rate or NPV < 0.

Income Statement

Income Statement		Current Period:
Revenue:	$	**1,300**
(-) Cost of Goods Sold (COGS):		(100)
Gross Profit:		**1,200**
(-) Cash Operating Expenses:		(800)
(-) Depreciation:		(100)
(-) Lease Depreciation:		(100)
(-) Stock-Based Compensation:		(50)
Total Operating Expenses:		**(1,050)**
Operating Income:		**150**
(+) Interest Income:		50
(-) Interest Expense:		(20)
(+) Gain / (-) Loss on PP&E Sale:		10
Pre-Tax Income:		**190**
(-) Income Tax Provision:		(48)
Net Income:		**143**
Common Shares (MM):		1,000
Earnings Per Share (EPS):	$	0.14

Shows a company's revenue, expenses, and taxes over a **PERIOD**, such as 1 year, 1 quarter, or 1 month; items must correspond to **delivery or allocation in the period** and affect the income available to the common shareholders.

Common Lines: Revenue, Cost of Goods Sold (COGS), Operating Expenses, Depreciation & Amortization, Gains/Losses, Interest Income/Expense, and Taxes.

Summary Lines: Gross Profit (Revenue – COGS), Operating Income (Revenue – COGS – OpEx), and Net Income (OpInc +/- Other Income/Expense – Taxes).

Balance Sheet

Balance Sheet		Current Period:
ASSETS:		
Current Assets:		
Cash:	$	100
Accounts Receivable:		100
Prepaid Expenses:		100
Inventory:		100
Total Current Assets:		**400**
Long-Term Assets:		
Net Property, Plant & Equipment (PP&E):		1,200
Finance Lease Assets:		200
Net Deferred Tax Asset (DTA):		100
Total Long-Term Assets:		**1,500**
Total Assets:	$	**1,900**
LIABILITIES & EQUITY:		
Current Liabilities:		
Accounts Payable:	$	300
Accrued Expenses:		200
Deferred Revenue:		200
Total Current Liabilities:		**700**
Long-Term Liabilities:		
Debt:		500
Finance Lease Liabilities:		200
Total Long-Term Liabilities:		**700**
Total Liabilities:		**1,400**
Common Shareholders' Equity:		**500**
Total Liabilities & Equity:	$	**1,900**
Balance Sheet Balanced?		**OK!**

Provides a snapshot of a company's **Assets** (its resources) and **Liabilities and Equity** (its funding sources) at a specific point in time; Assets must always equal Liabilities + Equity.

Assets: Can be sold for cash or provide a future economic benefit. Cash, Investments, Accounts Receivable, Inventory, Prepaid Expenses, Net PP&E, Goodwill, Other Intangibles, Lease Assets, and Deferred Tax Assets.

Liabilities: Funding sources and future obligations / cash outflows. Accounts Payable, Accrued Liabilities, Deferred Revenue, Lease Liabilities, Pensions Liabilities, and Debt.

Equity: Additional funding sources related to external parties (stock issuances) or internally generated via after-tax profits.

Cash Flow Statement

Cash Flow Statement		Current Period:
CASH FLOW FROM OPERATING ACTIVITIES:		
Net Income:	$	**143**
Non-Cash Expenses & Other Adjustments:		
(+) Depreciation:		100
(+) Lease Depreciation:		100
(+) Stock-Based Compensation:		50
(+/-) Deferred Income Taxes:		(13)
(-) Gain / (+) Loss on PP&E Sale:		(10)
Changes in Operating Assets & Liabilities:		
(+/-) Accounts Receivable:		(30)
(+/-) Prepaid Expenses:		(10)
(+/-) Inventory:		(15)
(+/-) Accounts Payable:		15
(+/-) Accrued Expenses:		20
(+/-) Deferred Revenue:		15
Cash Flow from Operations:		**365**
CASH FLOW FROM INVESTING ACTIVITIES:		
(-) Capital Expenditures:		(200)
(-) Additions to Finance Lease Ass		(50)
(+) PP&E Sale Proceeds:		50
Cash Flow from Investing:		**(200)**
CASH FLOW FROM FINANCING ACTIVITIES:		
(+) Issue Debt:		50
(-) Repay Debt:		(30)
(+) Issue Finance Leases:		10
(-) Repay Finance Leases:		(5)
(-) Common Dividends Issued:		(50)
(+) Issue New Common Shares:		10
(-) Repurchase Common Shares:		(20)
Cash Flow from Financing:		**(35)**
Beginning Cash:		**100**
Increase / (Decrease) in Cash:		**130**
Ending Cash:	$	**230**

Shows a company's **cash inflows and outflows over a period**; items may be adjustments to IS line items or additional cash inflows/outflows not on the IS.

Cash Flow from Operations: Net Income, Non-Cash Adjustments, and changes to Current Assets and Liabilities, excluding Debt and Cash.

Cash Flow from Investing: Changes to many Long-Term Assets and some non-operational Current Assets.

Cash Flow from Financing: Changes to many Long-Term Liabilities and Equity lines; also, some non-operational Current Liabilities (e.g., Short-Term Debt).

Asset **UP / DOWN** → Cash Flow **DOWN / UP**
Liability **UP / DOWN** → Cash Flow **UP / DOWN**

Accounting Interview Questions

Record **Revenue** and **Expenses** only when the product/service is **delivered or allocated**. If there's **no delivery/allocation**, use BS line items like Deferred Revenue and Prepaid Expenses for tracking.

Ending AR = Beginning AR + Non-Cash Rev. – Cash Collections of Rev.

Accounting Interview Questions

1) START with the **Income Statement** and explain how/if it changes.

2) Then, explain how the **Cash Flow Statement** changes, if at all (including changes in Deferred Taxes).

3) Finish by explaining how the **Balance Sheet** changes and why it still balances, i.e., why Assets still equal Liabilities + Equity. Make sure you explain changes in Cash and Equity.

Ending Inventory = Beginning Inventory + Inventory Purchases – Inventory Sold/Delivered and Recorded in COGS.

Ending Net PP&E = Beginning Net PP&E + CapEx – Depreciation – Losses + Gains – Sale Proceeds and Write-Downs.

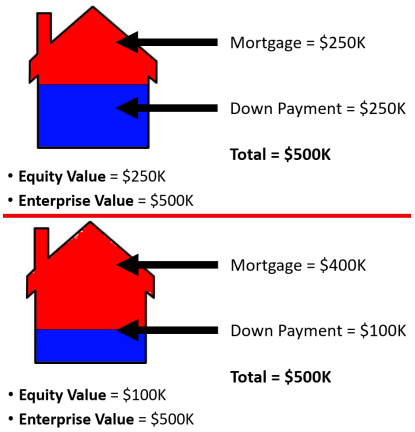
Ending AP = Beginning AP + Delivered-But-Unpaid-in-Cash Expenses – Cash Payment of Prior Expenses.

Ending DR = Beginning DR + Cash Collected for Undelivered Sales – Revenue Recognized for Previously Delivered Sales.

Ending Debt = Beginning Debt + Issuances – Repayments.

Ending Common Equity = Beginning Common Equity + Net Income + SBC – Dividends +/- Common Shares Issued/Repurchased.

Equity Value & Enterprise Value



Equity Value is the value of **EVERYTHING** a company has (Net Assets, or Total Assets – Total Liabilities), but only to the **EQUITY INVESTORS** (the common shareholders).

Enterprise Value is the value of the company's **CORE BUSINESS OPERATIONS** (Net Operating Assets, or Operating Assets – Operating Liabilities), but to **ALL INVESTORS** (Equity, Debt, Preferred, and possibly others).

Enterprise Value = Equity Value – Cash + Debt + Preferred Stock + Noncontrolling Interests.
(**More Advanced:** Subtract Investments, Non-Core Assets, and NOLs; add Unfunded Pensions, other funding sources, and maybe Leases.)

Diluted Equity Value = Share Price * (Basic Shares + Dilutive Shares from Options, Warrants, RSUs, Convertibles, etc.)

Treasury Stock Method: If the Options are in the money, Diluted Shares = # Options – # Options * Exercise Price / Current Share Price. Add these to the Basic Shares when calculating the Diluted Shares.

Does Common Shareholders' Equity (CSE) Change? If so, then Equity Value changes by that amount. If not, it does not change.

Do Net Operating Assets (NOA) Change? If so, then Enterprise Value (TEV) changes by that amount. If not, it does not change.

EX: A company issues $100 of common stock to buy a factory. CSE is up by $100, so Equity Value is up by $100. The factory is in Net PP&E, a core operating asset, so NOA is up by $100, and TEV is by $100.

Valuation Methodologies and Multiples

- **Company Value** =

$$\frac{\text{Cash Flow}}{\text{(Discount Rate – Cash Flow Growth Rate)}}$$

Where the Cash Flow Growth Rate Must Be < Discount Rate

- **Valuation is more than this formula because the Discount Rate and Cash Flow Growth Rate *change* over time**

You **value a company** based on its future cash flows (DCF Analysis) or the **multiples** from Comparable Public Companies and Precedent Transactions. A "multiple" is like a metric such as $ / square foot or $ / square meter when buying a home; it *normalizes* company values given their different sizes.

Screen for comparable companies based on geography, industry, and size (e.g., U.S. steel manufacturers with > $1 billion in revenue); for precedent transactions, add **time** (e.g., deals over the past 10 years).

To **calculate** valuation multiples, use **Equity Value** in the numerator if the denominator *deducts* the Interest Expense (e.g., Net Income and Free Cash Flow). Use **Enterprise Value** if it does not.

Metric:	Revenue	EBITDA	Net Income
Multiple:	TEV / Revenue	TEV / EBITDA	P / E
Calculation:	Enterprise Value / Revenue	Enterprise Value / (Operating Income + D&A on CFS + Non-Recurring Charges)	Equity Value / Net Income to Common
OpEx:	Not deducted	Deducted	Deducted
CapEx:	Not deducted	Not deducted	Partial deduction due to D&A
Interest/Taxes:	Not deducted	Not deducted	Deducted
Metric Meaning:	Net sales before expenses, taxes, and capital structure	Core, recurring business profitability before capital structure / taxes	Profit after taxes, capital structure, and non-core activities
Use Cases:	Generic for all/most companies; more useful for unprofitable/CF-negative ones	Normalize companies with different CapEx, D&A, capital structures, taxes, etc.	Quick/simple comparison but not useful for normalizing; industry-specific (FIG)

Basic Valuation: The median TEV / EBITDA in a set of comparable public companies is 10x. Your company's multiple is 8x. The comparable companies are in the same industry with similar revenue, growth rates, and margins → Therefore, your company "should be" trading at a 10x multiple, meaning its Enterprise Value should be ~25% higher. You can back into Equity Value from here by adding Cash, subtracting Debt, etc., and divide by its share count to determine its implied share price.

Public Comps are useful because they're based on real-world, near-term data, but there may not be truly comparable companies; **Precedent Transactions** are useful because they're based on the real prices paid for other companies, but the data is often spotty/misleading, and the implied values tend to be higher due to the **control premiums.**

Discounted Cash Flow (DCF) Analysis

Forecast Period Discount Rate:		10.0%
Initial Cash Flow in Forecast:		$ 100
PV of Cash Flows, Years 1 - 10:		819
Terminal Period Discount Rate:		9.0%
Terminal Cash Flow Growth Rate:		2.0%
PV of Cash Flows, Beyond Year 10:		2,467
Discounted to Present Value Today:		951
Implied Company Value:		**$ 1,770**

	Year 1	Year 2	Year 3
Annual Cash Flow:	$ 100	$ 110	$ 120
Cash Flow Growth Rate:		10.0%	9.0%

A DCF values a company based on the **Present Value of its future cash flows** over the next 5 – 10 years *plus* their **Present Value after that, in the "Terminal Period"** (whose value you estimate with the Terminal Value).

You almost always use an **Unlevered DCF**, which uses Unlevered Free Cash Flow and the Weighted Average Cost of Capital (WACC) for the Discount Rate. **Terminal Value** could be based on the Multiples Method (multiply the final forecast year EBITDA by a multiple from the comps) or the Gordon Growth Method (see right).

Terminal Value =

$$\frac{\text{Final Year FCF} * (1 + \text{FCF Growth Rate})}{(\text{Discount Rate} - \text{FCF Growth Rate})}$$

Unlevered FCF = Operating Income * (1 – Tax Rate) + D&A +/- Change in Working Capital – CapEx
(**More Advanced:** There may be adjustments for Deferred Taxes, Acquisitions, Intangible Purchases, Capitalized R&D, and Non-Recurring Charges. Always **deduct** the full Lease Expense for simplicity.)

ALWAYS EXCLUDE Interest Expense/Income, Preferred Dividends, Debt Principal Repayments, and Stock Issuances/Repurchases.

Like EBITDA, UFCF is **not** affected by the company's capital structure (Debt, Equity, Interest, etc.). However, the Discount Rate (WACC) *is* affected by capital structure, so the DCF output could still change.

Implied Enterprise Value in a DCF = Present Value of UFCF in Explicit Forecast Period + Present Value of Terminal Value → Then, add Cash, subtract Debt, etc., to get the Implied Equity Value.

DCF Type:	Unlevered	Levered
Metric:	Unlevered FCF (defined above)	LFCF = Net Income to Common + D&A +/- Change in Working Capital – CapEx +/- Net Change in Debt
Discount Rate:	WACC	Cost of Equity
Terminal Value (TV):	TEV / EBITDA for Terminal Multiple	P / E or Eq Val / LFCF for Terminal Multiple
Enterprise Value "Bridge":	Add Cash, subtract Debt, etc., to move to implied Eq Val	No bridge; The PV of LFCFs + PV of TV equals the Implied Equity Value
Valuation Output:	Results are often close but will **rarely, if ever, be the same** – very difficult to pick "equivalent" assumptions for real-life companies	
Use Cases:	Virtually all industry sectors outside of FIG (banks/insurance)	Some equity REITs and midstream oil & gas companies

Discount Rate (WACC)

The **Weighted Average Cost of Capital (WACC)** in a DCF represents the average annualized return you might expect to earn by investing in the entire capital structure (Equity, Debt, Preferred, etc.) of similar companies in the market. To a company, it represents the cost of funding its operations via **all** its sources of capital.

Metric:	Cost of Equity	Cost of Debt	WACC
Calculation:	Risk-Free Rate + Equity Risk Premium * Levered Beta (typically, the *re-levered Beta*)	Use the Yield to Maturity (YTM) on existing Debt; could also use a simple interest rate or Risk-Free Rate + Credit Default Spread	Cost of Equity * % Equity + Cost of Debt * (1 – Tax Rate) * % Debt + Cost of Preferred Stock * % Preferred Stock
Corresponds To:	Equity Value	Debt	Enterprise Value
Meaning:	Average annualized return you would expect from investing, including both income and price changes		
Smaller Co.:	Higher		
No Debt to Some Debt:	Higher	Higher	Lower, then Higher
Lower RFR:	Lower		
Lower ERP:	Lower	N/A	Lower
Lower Tax Rate:	Higher IF the company has Debt (if not, no direct impact)		

In Cost of Equity, you normally "un-lever" Beta for each comparable company, take the median, and then *re-lever* it based on the subject company's capital structure:

- **Unlevered Beta** = Levered Beta / (1 + Debt / Equity * (1 – Tax Rate) + Preferred Stock / Equity)
- **(Re-)Levered Beta** = Unlevered Beta * (1 + Debt / Equity * (1 – Tax Rate) + Preferred Stock / Equity)

Equity is **more expensive** than Debt because its risks and potential returns are both higher; also, the interest paid on Debt is tax-deductible, which makes the cash costs cheaper for the company.

M&A and Merger Models

Merger Model Walkthrough: 1) Project Buyer/Seller statements.

2) Estimate the Purchase Price and Cash, Debt, and Stock funding.

3) Create Sources & Uses and Purchase Price Allocation schedules.

4) Combine the Balance Sheets and reflect Cash, Debt, and Stock used, new Goodwill, and Seller's Common Equity write-down.

5) Combine Income Statements, including Foregone Interest on Cash, Interest on New Debt, and Synergies.

6) Combined Net Income = Combined Pre-Tax Income * (1 – Buyer's Tax Rate); Combined EPS = Combined Net Income / (Buyer's Shares + Shares Issued in Deal)

7) EPS Accretion / Dilution = Combined EPS / Buyer's Standalone EPS – 1

M&A and Merger Models

Deals are **accretive** when the Combined EPS is higher than the Buyer's standalone EPS and **dilutive** when it's lower:

Component:	Calculation
Cost of Cash:	Foregone Interest Rate on Cash * (1 – Buyer's Tax Rate)
Cost of Debt:	Interest Rate on New Debt * (1 – Buyer's Tax Rate)
Cost of Stock:	*Buyer's* Net Income / *Buyer's* Equity Value
Weighted Cost of Acquisition:	% Cash Used * Cost of Cash + % Debt Used * Cost of Debt + % Stock Used * Cost of Stock
Seller's Purchase Yield:	Seller's Net Income / Purchase Equity Value
Accretive/Dilutive?	Accretive if Weighted Cost < Purchase Yield; Dilutive if Weighted Cost > Purchase Yield

If the deal is **100% Stock**, compare the Buyer's P / E multiple to the Seller's Purchase P / E multiple. If the Buyer's multiple is higher, the deal will be accretive; if not, it will be dilutive.

M&A Purchase Price: Based on a premium to the Seller's current share price for public companies; for private companies, use a multiple such as TEV / EBITDA. The "true price" is the Purchase Enterprise Value.

Financing Mix: Buyers use as much Cash as possible first, then Debt, and then Stock because of their respective costs. Cash is limited based on Min. Cash requirements, and Debt is normally limited based on a max or targeted Debt / EBITDA.

Sources & Uses: Demonstrates where the funding comes from and where it goes to; existing Debt is almost always refinanced and replaced with a new issuance of the same size (terms may differ).

Sources & Uses of Funds:

Sources:		Uses:	
Cash Used:	$ 40.0	Equity Purchase Price of Target:	$ 990.0
New Debt Issued:	220.0	Assume/Replace Target's Debt:	250.0
Stock Issued:	730.0	Transaction Fees:	14.9
Assume/Replace Target's Debt:	250.0	Financing Fees:	9.4
Cash for Transaction Fees:	24.3	**Total Uses:**	**$ 1,264.3**
Total Sources:	**$ 1,264.3**		
		CHECK:	**OK!**

Combined Metric*:	Calculation	Expected Range
Combined Equity Value	Buyer's Equity Value + Market Value of Stock Issued	Buyer's Eq. Val. to Buyer's Eq. Val. + Seller's Purchase Eq. Val
Combined Enterprise Value (TEV)	Buyer's Enterprise Value + Seller's Purchase Enterprise Value	Buyer's TEV + Seller's Purchase TEV, but often falls post-announcement
Combined TEV Multiples	Combined TEV / (Combined Revenue, EBITDA, etc. of Buyer and Seller)	Must be between Buyer's multiples and Seller's purchase multiples
Combined Eq. Val Multiples	Must recalculate manually based on Combined Eq Val and Net Income, including acquisition effects	*Normally* between Buyer's multiples and Seller's purchase multiples, but do not have to be

*Everything may change based on market's reaction to deal; Buyer's share price often falls (for example).

Leveraged Buyouts and LBO Models

LBO Model Walkthrough: 1) Assume a Purchase Price, Debt and Equity used, Interest Rate on Debt, and Operational Drivers (revenue growth, margins, etc.).

2) Create a Sources & Uses schedule to reflect the Debt and Investor Equity used, Fees, and treatment of the company's Cash and Debt.

3) Project the company's Income Statement and a partial Cash Flow Statement down to the Free Cash Flow (FCF) line.

4) CF Avail. For Debt Repayment = Beginning Cash + FCF – Minimum Cash (and subtract Mandatory Debt Repayments if applicable).

5) Repay this amount of Debt each year (and the mandatory repayments) and link the Interest Expense to this changing Debt balance, ensuring that FCF also deducts the Interest.

6) Assume an EBITDA exit multiple, back into the Exit Equity Value, and calculate the IRR and MoM multiple based on these Exit Equity Proceeds vs. the Investor Equity in the beginning.

Ideal LBO Candidates: Stable / predictable cash flows, modest CapEx, fragmented industry, growth / margin opportunities, strong management, solid asset bases, and realistic paths to exit.

Debt Types: Revolver (temporary borrowing), Secured Debt (Lower floating rates, principal repayments, collateral, maintenance covenants), and Unsecured Debt (Higher fixed rates, no principal repayments, and incurrence covenants). Draw on Revolver when CF Avail. For Debt Repayment turns negative and repay when it's positive.

Returns Sources: Multiple Expansion (buy for 10x EBITDA and sell for 12x), EBITDA Growth, and Debt Paydown / Cash Generation.

Returns Drivers: Purchase and Exit Assumptions matter most; % Debt used is very significant; revenue growth, margin, and cash flow assumptions are also important. Specific Debt terms matter less.

LBO Valuation: Use Goal Seek to solve for the *maximum price* a PE firm could pay to achieve its targeted IRR, such as 20%, with the operational, exit, and debt / equity assumptions as "constants."

LBO Math (for "in between" values, approximate via averages):

Multiple and Time Frame:	Approximate IRR:
2x in 3 Years	~25%
2x in 5 Years	~15%
3x in 3 Years	~45%
3x in 5 Years	~25%

EX: Buy a $100 EBITDA company for 10x EBITDA and 6x Debt / EBITDA, sell at $150 EBITDA for 9x in 5 years, and repay $250 of Debt. Exit TEV = $150 *9 = $1,350. Subtract the $600 – $250 = $350 of remaining Debt. $1,000 Exit Equity Proceeds / $400 Investor Equity = 2.5x, so it's a ~20% IRR over 5 years since 2.5x is right between 2x (~15%) and 3x (~25%).